

meeting format and instructions, a summary of the aggregated responses from round 2 of the Delphi survey (supplementary table 3), and the draft TRIPOD+AI checklist. The checklist circulated to the consensus meeting participants included 59 items covering: title (one item), abstract (one item), introduction (four items), methods (32 items), results (11 items), discussion (eight items), and other (two items).

Given the high endorsement achieved for many items in round 2, a subset of 17 items were highlighted for plenary discussion and voting during the consensus meeting. After discussion, participants were given one minute to vote to include or exclude the item from the TRIPOD+AI checklist. The voting was registered using

the poll function of the online meeting program. The 17 items included one item that had not achieved consensus in round 2 and 16 items that had undergone rewording after round 2 or were new items that were not included in TRIPOD 2015. After discussion and voting on these 17 items, the final TRIPOD+AI checklist was formed.

TRIPOD+AI statement

TRIPOD+AI comprises a checklist of items that are considered essential for good reporting of studies developing or evaluating (validating) a prediction model using any statistical or machine learning methods (table 2). Box 2 summarises noteworthy additions and changes to TRIPOD 2015. The TRIPOD+AI checklist comprises

Table 2 | TRIPOD+AI checklist for the reporting of prediction model studies

Section/topic	Item	Development/evaluation*	Checklist item
Title			
Title	1	D;E	Identify the study as developing or evaluating the performance of a multivariable prediction model, the target population, and the outcome to be predicted
Abstract			
Abstract	2	D;E	See TRIPOD+AI for Abstracts checklist
Introduction			
Background	3a	D;E	Explain the healthcare context (including whether diagnostic or prognostic) and rationale for developing or evaluating the prediction model, including references to existing models
	3b	D;E	Describe the target population and the intended purpose of the prediction model in the context of the care pathway, including its intended users (eg, healthcare professionals, patients, public)
	3c	D;E	Describe any known health inequalities between sociodemographic groups
Objectives	4	D;E	Specify the study objectives, including whether the study describes the development or validation of a prediction model (or both)
Methods			
Data	5a	D;E	Describe the sources of data separately for the development and evaluation datasets (eg, randomised trial, cohort, routine care or registry data), the rationale for using these data, and representativeness of the data
	5b	D;E	Specify the dates of the collected participant data, including start and end of participant accrual; and, if applicable, end of follow-up
Participants	6a	D;E	Specify key elements of the study setting (eg, primary care, secondary care, general population) including the number and location of centres
	6b	D;E	Describe the eligibility criteria for study participants
	6c	D;E	Give details of any treatments received, and how they were handled during model development or evaluation, if relevant
Data preparation	7	D;E	Describe any data pre-processing and quality checking, including whether this was similar across relevant sociodemographic groups
Outcome	8a	D;E	Clearly define the outcome that is being predicted and the time horizon, including how and when assessed, the rationale for choosing this outcome, and whether the method of outcome assessment is consistent across sociodemographic groups
	8b	D;E	If outcome assessment requires subjective interpretation, describe the qualifications and demographic characteristics of the outcome assessors
	8c	D;E	Report any actions to blind assessment of the outcome to be predicted
Predictors	9a	D	Describe the choice of initial predictors (eg, literature, previous models, all available predictors) and any pre-selection of predictors before model building
	9b	D;E	Clearly define all predictors, including how and when they were measured (and any actions to blind assessment of predictors for the outcome and other predictors)
	9c	D;E	If predictor measurement requires subjective interpretation, describe the qualifications and demographic characteristics of the predictor assessors
Sample size	10	D;E	Explain how the study size was arrived at (separately for development and evaluation), and justify that the study size was sufficient to answer the research question. Include details of any sample size calculation
Missing data	11	D;E	Describe how missing data were handled. Provide reasons for omitting any data
Analytical methods	12a	D	Describe how the data were used (eg, for development and evaluation of model performance) in the analysis, including whether the data were partitioned, considering any sample size requirements
	12b	D	Depending on the type of model, describe how predictors were handled in the analyses (functional form, rescaling, transformation, or any standardisation)
	12c	D	Specify the type of model, rationale, all model building steps, including any hyperparameter tuning, and method for internal validation
	12d	D;E	Describe if and how any heterogeneity in estimates of model parameter values and model performance was handled and quantified across clusters (eg, hospitals, countries). See TRIPOD-Cluster for additional considerations†
	12e	D;E	Specify all measures and plots used (and their rationale) to evaluate model performance (eg, discrimination, calibration, clinical utility) and, if relevant, to compare multiple models
	12f	E	Describe any model updating (eg, recalibration) arising from the model evaluation, either overall or for particular sociodemographic groups or settings
	12g	E	For model evaluation, describe how the model predictions were calculated (eg, formula, code, object, application programming interface)

(Continued)

Table 2 | Continued

Section/topic	Item	Development/evaluation*	Checklist item
Class imbalance	13	D;E	If class imbalance methods were used, state why and how this was done, and any subsequent methods to recalibrate the model or the model predictions
Fairness	14	D;E	Describe any approaches that were used to address model fairness and their rationale
Model output	15	D	Specify the output of the prediction model (eg, probabilities, classification). Provide details and rationale for any classification and how the thresholds were identified
Training versus evaluation	16	D;E	Identify any differences between the development and evaluation data in healthcare setting, eligibility criteria, outcome, and predictors
Ethical approval	17	D;E	Name the institutional research board or ethics committee that approved the study and describe the participant informed consent or the ethics committee waiver of informed consent
Open science			
Funding	18a	D;E	Give the source of funding and the role of the funders for the present study
Conflicts of interest	18b	D;E	Declare any conflicts of interest and financial disclosures for all authors
Protocol	18c	D;E	Indicate where the study protocol can be accessed or state that a protocol was not prepared
Registration	18d	D;E	Provide registration information for the study, including register name and registration number, or state that the study was not registered
Data sharing	18e	D;E	Provide details of the availability of the study data
Code sharing	18f	D;E	Provide details of the availability of the analytical code§
Patient and public involvement			
Patient and public involvement	19	D;E	Provide details of any patient and public involvement during the design, conduct, reporting, interpretation, or dissemination of the study or state no involvement
Result			
Participants	20a	D;E	Describe the flow of participants through the study, including the number of participants with and without the outcome and, if applicable, a summary of the follow-up time. A diagram may be helpful
	20b	D;E	Report the characteristics overall and, where applicable, for each data source or setting, including the key dates, key predictors (including demographics), treatments received, sample size, number of outcome events, follow-up time, and amount of missing data. A table may be helpful. Report any differences across key demographic groups
	20c	E	For model evaluation, show a comparison with the development data of the distribution of important predictors (demographics, predictors, and outcome)
Model development	21	D;E	Specify the number of participants and outcome events in each analysis (eg, for model development, hyperparameter tuning, model evaluation)
Model specification	22	D	Provide details of the full prediction model (eg, formula, code, object, application programming interface) to allow predictions in new individuals and to enable third party evaluation and implementation, including any restrictions to access or reuse (eg, freely available, proprietary)¶
Model performance	23a	D;E	Report model performance estimates with confidence intervals, including for any key subgroups (eg, sociodemographic). Consider plots to aid presentation
	23b	D;E	If examined, report results of any heterogeneity in model performance across clusters. See TRIPOD-Cluster for additional details‡
Model updating	24	E	Report the results from any model updating, including the updated model and subsequent performance
Discussion			
Interpretation	25	D;E	Give an overall interpretation of the main results, including issues of fairness in the context of the objectives and previous studies
Limitations	26	D;E	Discuss any limitations of the study (such as a non-representative sample, sample size, overfitting, missing data) and their effects on any biases, statistical uncertainty, and generalisability
Usability of the model in the context of current care	27a	D	Describe how poor quality or unavailable input data (eg, predictor values) should be assessed and handled when implementing the prediction model
	27b	D	Specify whether users will be required to interact in the handling of the input data or use of the model, and what level of expertise is required of users
	27c	D;E	Discuss any next steps for future research, with a specific view to applicability and generalisability of the model

TRIPOD=Transparent Reporting of a multivariable prediction model for Individual Prognosis Or Diagnosis; AI=artificial intelligence.

*D=items relevant only to the development of a prediction model; E=items relating solely to the evaluation of a prediction model; D;E=items applicable to both the development and evaluation of a prediction model.

†Separately for all model building approaches.

‡TRIPOD-Cluster is a checklist of reporting recommendations for studies developing or validating models that explicitly account for clustering or explore heterogeneity in model performance (eg, at different hospitals or centres).^{19 20}

§Relates to the analysis code, for example, any data cleaning, feature engineering, model building, and evaluation.

¶Relates to the code to implement the model to get estimates of risk for a new individual.

27 main items about the title (item 1), abstract (item 2), introduction (items 3 and 4), methods (items 5-17), open science practises (item 18), patient and public involvement (item 19), results (items 20-24), and discussion (items 25-27). Some items included multiple subitems, totalling to 52 checklist subitems.

TRIPOD+AI covers studies that describe the development of a prediction model, the evaluation (validation) of prediction model performance, or both. Any items denoted D;E apply to all studies regardless

of whether they are developing a prediction model or evaluating the performance of a prediction model (table 2). Items in the checklist denoted D apply to studies that describe the development of a prediction model, while items denoted E apply to studies that evaluate the performance of a prediction model. For studies both developing and evaluating the performance of a prediction model, all checklist items apply.

A separate checklist for journal or conference abstracts of prediction model studies is included in

Box 2: Noteworthy changes and additions to TRIPOD 2015

- New checklist of reporting recommendations to cover prediction model studies using any regression or machine learning method (eg, random forests, deep learning), and harmonise nomenclature between regression and machine learning communities
- New TRIPOD+AI checklist supersedes the TRIPOD 2015 checklist, which should no longer be used
- Particular emphasis on fairness (box 1) to raise awareness and ensure that reports mention whether specific methods were used to deal with fairness. Aspects of fairness are embedded throughout the checklist
- Inclusion of TRIPOD+AI for Abstracts for guidance on reporting abstracts
- Modification of the model performance item recommending that authors evaluate model performance in key subgroups (eg, sociodemographic)
- Inclusion of a new item on patient and public involvement to raise awareness and prompt authors to provide details on any patient and public involvement during the design, conduct, reporting (and interpretation), and dissemination of the study
- Inclusion of an open science section with subitems on study protocols, registration, data sharing and code sharing

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TRIPOD+AI. This checklist updates the TRIPOD for Abstracts statement,¹⁸ reflecting new content and maintaining consistency with TRIPOD+AI (table 3).

The recommendations in TRIPOD+AI are for transparently reporting how prediction model research was conducted; it does not prescribe how to develop or evaluate a prediction model. The checklist is not a quality appraisal tool. Readers are referred to PROBAST^{90 91} and the forthcoming PROBAST+AI⁷⁹ to assess the quality and risk of bias of prediction models (<https://www.probast.org/>).

How to use TRIPOD+AI

The TRIPOD+AI checklist supersedes the TRIPOD 2015 checklist, which should no longer be used. For prediction model studies that have accounted

for clustering (eg, multiple hospitals, multiple datasets), authors should consult TRIPOD-Cluster for additional reporting recommendations.^{19 20} The 2015 explanation and elaboration document remains an important document to provide background and examples for most of the TRIPOD+AI reporting items¹⁷ (because many items have not changed or have been minimally changed), while we produce a detailed and updated document for TRIPOD+AI. We recommend using TRIPOD+AI early in the writing process to ensure that all key details are addressed and reported. An expanded checklist in a bullet point structure has been developed (supplementary table 1) to facilitate implementation of TRIPOD+AI by providing a brief rationale and guidance for each item in the checklist.

Although many of the items in the TRIPOD+AI checklist have a natural order and sequence in a report, some do not. We do not stipulate a structured format or dictate where each individual reporting recommendation should appear in a prediction model report or publication, because this order might also depend on journal formatting policies.

The recommendations contained within TRIPOD+AI are the minimum reporting recommendations, and authors may provide additional information. If journal word limits and restrictions on number of tables and figures in the main body of the manuscript complicate reporting, authors can report and reference some of the requested or additional information in supplementary material. If the information required is already reported in a publicly accessible study protocol, then referring to that document may suffice. If a particular checklist item cannot be discussed in the report because the information is unknown or irrelevant, then this should be acknowledged and clearly stated. Additional files and study materials not included in the supplementary material should be deposited in

Table 3 | Essential items to include for the reporting of prediction model studies in a journal or conference abstract (TRIPOD+AI for Abstracts*)

Section and item	Checklist item
Title	
1	Identify the study as developing or evaluating the performance of a multivariable prediction model, the target population, and the outcome to be predicted
Background	
2	Provide a brief explanation of the healthcare context and rationale for developing or evaluating the performance of all models
Objectives	
3	Specify the study objectives, including whether the study describes model development, evaluation, or both
Methods	
4	Describe the sources of data
5	Describe the eligibility criteria and setting where the data were collected
6	Specify the outcome to be predicted by the model, including time horizon of predictions in case of prognostic models
7	Specify the type of model, a summary of the model-building steps, and the method for internal validation†
8	Specify the measures used to assess model performance (eg, discrimination, calibration, clinical utility)
Results	
9	Report the number of participants and outcome events
10	Summarise the predictors in the final model†
11	Report model performance estimates (with confidence intervals)
Discussion	
12	Give an overall interpretation of the main results
Registration	
13	Give the registration number and name of the registry or repository

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*This checklist is based on the TRIPOD for Abstracts statement published in 2020,¹⁷ but has been revised and updated for consistency with the TRIPOD+AI statement.

†Relevant only to studies describing the development of a prediction model.

general purpose (eg, Open Science Framework, Dryad, figshare) or institutional open access repositories that provide free access in perpetuity. Details of access to any additional files should be referenced and linked, for example, with a doi number in the main study report or publication.

We recommend that authors submit a completed checklist indicating the page or line where each requested item can be found, to help the editorial and peer review process. A template for the TRIPOD+AI checklist for filling out separately can be found in supplementary table 2 and is available to download from www.tripod-statement.org.

News, announcements, and information relating to TRIPOD+AI can be found on the TRIPOD website (www.tripod-statement.org) and on social media accounts such as X (formerly known as Twitter; @TRIPODStatement). The Enhancing the Quality and Transparency Of health Research (EQUATOR) Network (<https://www.equator-network.org/>) will also disseminate and promote the TRIPOD+AI statement. Translation of TRIPOD+AI into different languages is welcomed and encouraged, please contact the corresponding author. Translations should use the structured and predefined process that includes authors of the original publication and receives their approval. The TRIPOD website contains further details on translation (www.tripod-statement.org).

Discussion

TRIPOD+AI has been developed through an international multi-stakeholder consensus process. It provides minimum reporting recommendations for studies describing the development or evaluation (validation) of prediction models using any regression or machine learning methods. At the time of guideline development, foundation and large language models (such as ChatGPT) that are rapidly gaining momentum were not considered—the TRIPOD+AI guidance is primarily aimed at non-generative models. However, many of the principles are applicable for driving transparency in generative AI studies in health. Periodic updating of TRIPOD+AI will be needed to remain relevant and reflect advancements in AI and machine learning methods, for example, by explicitly looking at generative approaches.

TRIPOD+AI was developed by updating TRIPOD 2015, with recommendations informed by systematic reviews of the literature, a Delphi survey, and an online consensus meeting. Reporting TRIPOD+AI items can help users to understand and appraise the quality of the study methods, increasing transparency around the study findings, reducing overinterpretation of study findings, facilitating replication and reproducibility, and aiding implementation of the prediction model. The checklist items are minimum reporting recommendations, and authors will typically provide additional details on the data, study design, methods, analysis, results, and discussion.

TRIPOD+AI emphasises fairness issues throughout the checklist, which was lacking or not explicitly

stated in TRIPOD 2015.³³ Fairness in prediction model research is particularly important in healthcare, which has gained prominence with AI and machine learning methods being used to develop models to assist in decision making. Fairness in this context means that prediction models are designed and used in a way that does not adversely discriminate against any particular group of individuals and does not create or exacerbate (and ideally mitigates or reduces) existing inequalities in healthcare provision or patient outcomes.⁹² One important aspect of fairness is ensuring that the data used to develop or evaluate prediction models are representative and diverse, and that limitations of data bias are acknowledged, dealt with, and mitigated during model development. The STANDING Together initiative is in the process of developing standards for data diversity, inclusivity, and generalisability to tackle bias in AI health datasets.⁶²

Data should ideally include information from individuals of different ages, sexes/genders, and races/ethnicities, with different health conditions or comorbidities and from different geographical locations. These differences should be representative of the population in whom the prediction model is intended to be used. If the data used to develop the models do not adequately represent the full diversity of the intended use population, the resulting model might not perform as expected in those missing from the data, which should be clearly stated. If the data used to evaluate a model are not representative of the target population, then the estimated predictive accuracy in subgroups (eg, defined by relevant personal, social, or clinical attributes) could be biased and misleading.

While adequate representation of minoritised and underserved groups within datasets is one key element to achieving fairness goals, representation alone does not guarantee fairness.^{61 93} Therefore, TRIPOD+AI has embedded items on fairness throughout, including in the background (item 3c), methods (items 5a, 7a, 8a, 8b, 9c, 12f, 14), results (items 20b, 23a), and discussion (items 25, 26).

Fairness in healthcare also means involving diverse stakeholders, including patients, the general public and clinicians, in the development, evaluation, implementation, and deployment of a prediction model into the clinical pathway.⁹⁴ Involving a variety of perspectives will help to ensure that the prediction model is, in principle, designed to meet the needs of all individuals and is used in a way that is fair and equitable, promoting health equity. TRIPOD+AI includes item 19 on public and patient involvement to incentivise the integration of patient involvement in prediction model studies beyond a mere tick box exercise, to encourage and promote the principles of open science and engagement, and to ensure better clinical and public acceptability of the work.

TRIPOD+AI prominently features open science practices.³⁵ Open science practices are crucial for prediction model research in healthcare as they promote transparency, reproducibility, and collaboration between researchers.⁹⁵ By registering

research and making study materials such as protocols, data, code, and the prediction model open available, other researchers can verify the findings and evaluate model performance in new data to ensure that models are accurate, and evaluate models for safety. Open science practices also enable researchers to build on each other's work, leading to more efficient progress in healthcare. These practices can have a considerable impact on patient outcomes by improving the accuracy, integrity, and reliability of prediction models. If data are openly shared, clinicians and researchers can develop or evaluate models on larger and more diverse sets of patient data,⁹⁶ potentially leading to more accurate predictions and better informed decisions for patient healthcare. Therefore, TRIPOD+AI includes a section on open science, covering issues such as

funding declarations (item 18a), conflicts of interest (18b), protocol availability (18c), study registration (18d), data sharing (18e), and code sharing (18f).

We anticipate that the key users and beneficiaries of TRIPOD+AI will be researchers writing papers, journal editors and peer reviewers who evaluate research papers, and other stakeholders (eg, academic institutions, policy makers, funders, regulators, patients, study participants, and the broader public) who will benefit from the increased quality of prediction model research (table 4). The guideline is relevant for any reports related to clinical prediction model development and validation studies, including medical research articles and other areas where evidenced reports are needed, for example, to accompany software and tools.

Table 4 | Adherence to the TRIPOD+AI reporting guideline: potential benefits from stakeholders' actions

User/stakeholder	Proposed action	Potential benefits
Academic institutions	Promote or require adherence of TRIPOD+AI by investigators developing, evaluating, or implementing prediction models	Enhance a culture of transparency in the design, analysis, and reporting of prediction model research
	Provide training for early career researchers on the importance and benefits of transparent and complete reporting, including requiring doctoral students to write their thesis and manuscripts in accordance with the full TRIPOD+AI guideline	Improve the quality, accountability, reproducibility, replicability, and usefulness of produced research
Researchers	Adhere to TRIPOD+AI when writing studies for publication	Improved completeness and quality of reporting
		Increased awareness of the minimal detail required and expected when writing a prediction model publication
		Improved quality, accountability, reproducibility, replicability, and usefulness of produced research
		Improved reporting of details that facilitate independent evaluation of the model
Journal editors	Require and enforce authors to use TRIPOD+AI and submit a completed a checklist when writing the manuscript Recommend peer reviewers use TRIPOD+AI	Improved understanding of journal requirements and expectations for prediction model publications
		Increased efficiency of peer review resulting from improved author understanding of journal requirements for prediction model publications
		Improved quality, accountability, reproducibility, replicability, and usefulness of published research
		Improved reporting of details that facilitate independent evaluation of the model
Peer reviewers	Use TRIPOD+AI to evaluate completeness of reporting	Improve the efficiency and quality of peer review
		Facilitate and direct specific feedback to authors on where important details are missing
Funders	Recommend or mandate use of TRIPOD+AI by investigators when receiving a grant for prediction model research	Increase the usefulness of research outputs
		Reduce avoidable research waste due to incomplete reporting
		Ensure that funded research can be used by others
Patients, public, and study participants	Advocate use of TRIPOD+AI by authors, peer reviewers, journals, and funders	Improved trust in research findings
		Improved understanding of prediction model research
		Promote health equity considerations in research
		Align patient reported outcomes and patient experience with clinical research outcomes for precision medicine and personalised disease management
Systematic reviewers and meta-researchers	Use TRIPOD+AI to assess completeness of reporting Use TRIPOD+AI as an aid when assessing quality and risks of bias	Improved evaluation of study quality when used alongside risk of bias tools (eg, PROBAST)
		Improved availability of data needed for meta-analysis
Policy makers	Use or promote TRIPOD+AI to ensure research is transparently and completely reported	Ensure decisions to evaluate or implement a prediction model are based on complete and transparently reported information
		Add integrity for evidence based policy recommendations
Regulators	Clinical reviewers use TRIPOD+AI to assess completeness of clinical investigation reporting for "software as a medical device" regulatory submissions where the operating principle of the product is a prediction model	Align reported intended use with regulatory intended purpose
		Align medical device regulatory review and pivotal investigational reporting with best practice
		Encourage manufacturers to publish clinical investigation reports by encouraging one common standard
Technology and medical device manufacturers	Verify whether sufficient details about a model are available to enable development and manufacturing of technology and devices	Encourages manufacturers to publish clinical investigation reports by encouraging one common standard
Healthcare professionals	Verify whether sufficient details about a model are available before purchasing or using a model to support clinical use	Improved understanding of the target population of a model and the clinical decision it is intended to support
		Improved understanding of model predictions and awareness of limitations
		Improved trust in research findings

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